

Assignment-01

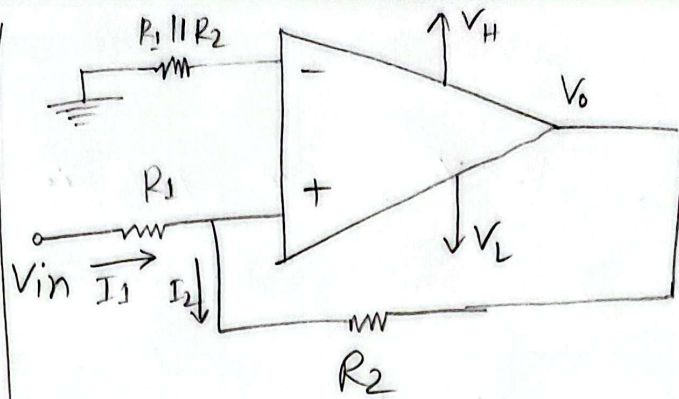
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1. Find the expression for V_H , V_L , V_T , V_- .
2. Draw the VTC.

Ans. to the que. no-01

from the circuit,

$$\boxed{V_- = 0V}$$

Now, KCL at V_+ ,

$$I_1 = I_2$$

$$\Rightarrow \frac{V_{in} - V_+}{R_1} = \frac{V_+ - V_o}{R_2}$$

$$\Rightarrow \frac{V_{in} - V_+}{R_1} - \frac{V_+ - V_o}{R_2} = 0$$

$$\Rightarrow \frac{V_{in}}{R_1} - \frac{V_+}{R_1} - \frac{V_+}{R_2} + \frac{V_o}{R_2} = 0$$

$$\Rightarrow (-V_+) \left(\frac{R_1 + R_2}{R_1 R_2} \right) = - \left(\frac{R_2 V_{in} + R_1 V_o}{R_1 R_2} \right)$$

$$\Rightarrow V_+ = \frac{R_2 V_{in} + R_1 V_o}{R_1 + R_2}$$

$$\Rightarrow \boxed{V_+ = \left(\frac{R_2}{R_1 + R_2} \right) V_{in} + \left(\frac{R_1}{R_1 + R_2} \right) V_o}$$

Now, if V_{in} is very large negative voltage, then

$$V_t < V_{th}, \text{ so, } V_o = V_L$$

Now, at $V_t = 0$, it will switch to V_H if V_{in} crosses V_{TH} .

$$\therefore V_t = 0$$
$$\Rightarrow \left(\frac{R_2}{R_1 + R_2} \right) V_{TH} + \left(\frac{R_1}{R_1 + R_2} \right) V_L = 0$$

$$\Rightarrow \left(\frac{R_2}{R_1 + R_2} \right) V_{TH} = - \left(\frac{R_1}{R_1 + R_2} \right) V_L$$

$$\Rightarrow \boxed{V_{TH} = - \frac{R_1}{R_2} V_L}$$

Again if V_{in} is very large positive voltage, then

$$V_t > V_{th} \text{ so, } V_o = V_H$$

so, at $V_t = 0$, it will switch to V_L if V_{in} crosses V_{TL} .

$$V_t = 0$$
$$\Rightarrow \left(\frac{R_2}{R_1 + R_2} \right) V_{TL} + \left(\frac{R_1}{R_1 + R_2} \right) V_H = 0$$

$$\Rightarrow \boxed{V_{TL} = - \frac{R_1}{R_2} V_H} \quad (\text{Ans.})$$

Ans. to the que no-02

